Pneumonia Diagnosis Detection and Localization

UE24CS352A Machine Learning Mini-Project (Problem 34) | Team: [Nakka Gopya , PES1UG24AM107] and [Dharshini V , PES1UG24AM428] | Section: [B]

1. Problem Statement

Given a chest X-ray, decide whether it shows pneumonia (detection) and highlight the region of the image that drives the decision (localization). Fast, consistent screening support matters because pneumonia is a leading cause of death in children.

2. Dataset

We used the Kaggle *Chest X-Ray Images (Pneumonia)* dataset (5,856 pediatric anteroposterior X-rays, classes NORMAL and PNEUMONIA). Its official validation set has only 16 images, so we pooled all images and re-split them 70/15/15 **grouped by patient ID** (parsed from filenames) to prevent the same patient appearing in two splits. The data is imbalanced (about 73% pneumonia).

Split	Images	NORMAL	PNEUMONIA
Train	4,077	1,061	3,016
Validation	901	258	643
Test	878	264	614

3. Approach

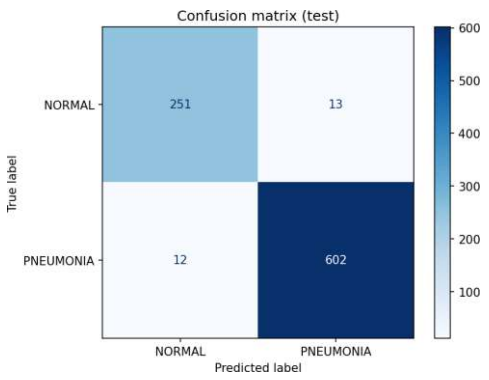
Classification: transfer learning with an ImageNet-pretrained DenseNet121 whose final layer is replaced by a 2-class linear layer, fine-tuned end to end. Dense connectivity reuses features well on small medical datasets. **Imbalance:** class-weighted cross-entropy (weights inversely proportional to class frequency). **Localization:** Grad-CAM, which weights the last convolutional feature maps by the gradient of the predicted class score and shows the result as a heatmap. It needs no bounding-box labels, which this dataset lacks.

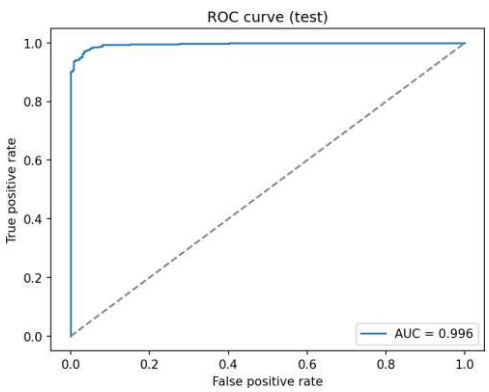
4. Implementation Overview

- Images resized to 224x224, ImageNet normalization; training augmentation: random resized crop, rotation up to 10 degrees, brightness/contrast jitter.
- AdamW (lr 1e-4, weight decay 1e-4), cosine learning-rate schedule, batch size 32, 6 epochs, mixed precision on a Colab T4 GPU. Best epoch chosen by validation ROC-AUC.
- Code: PyTorch/torchvision; `src/train.py` (split, train, evaluate), `src/common.py` (model, Grad-CAM), `app/app.py` (Gradio demo: upload X-ray, get label, confidence and heatmap).

5. Results (held-out test set, 878 images)

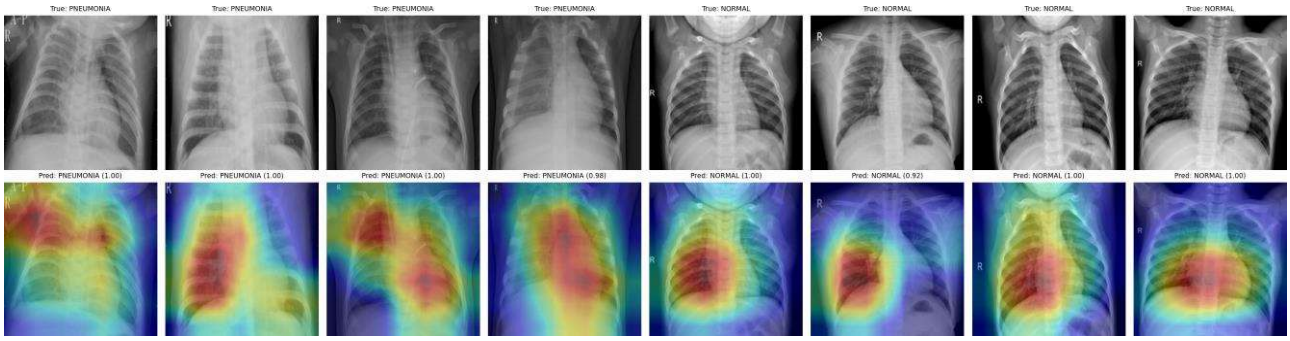
Class	Precision	Recall	F1	Support
NORMAL	0.954	0.951	0.953	264
PNEUMONIA	0.979	0.980	0.980	614
Accuracy			0.972	878
ROC-AUC			0.996	





Confusion matrix: 12 pneumonia cases missed, 13 normal cases flagged.

ROC curve, AUC = 0.996.



Grad-CAM on test images (top: X-ray, bottom: heatmap; red = most influential region).

6. Conclusions and Limitations

A fine-tuned DenseNet121 reaches 97.2% test accuracy and 0.996 AUC, with 98% recall on pneumonia, and Grad-CAM mostly highlights lung regions, which makes predictions easier to inspect. Limitations: (1) the data is pediatric and from one source, so results may not transfer to adults or other hospitals; (2) Grad-CAM is a coarse, qualitative localization that we did not score against ground-truth boxes, and some heatmaps touch image markers or borders, a sign of possible shortcut cues; (3) the Kaggle labels are known to contain some noise. Future work: train an object detector (Faster R-CNN or YOLO) on the RSNA bounding-box dataset, evaluate on external data, and calibrate the decision threshold toward higher recall. This is an educational prototype, not a medical device.